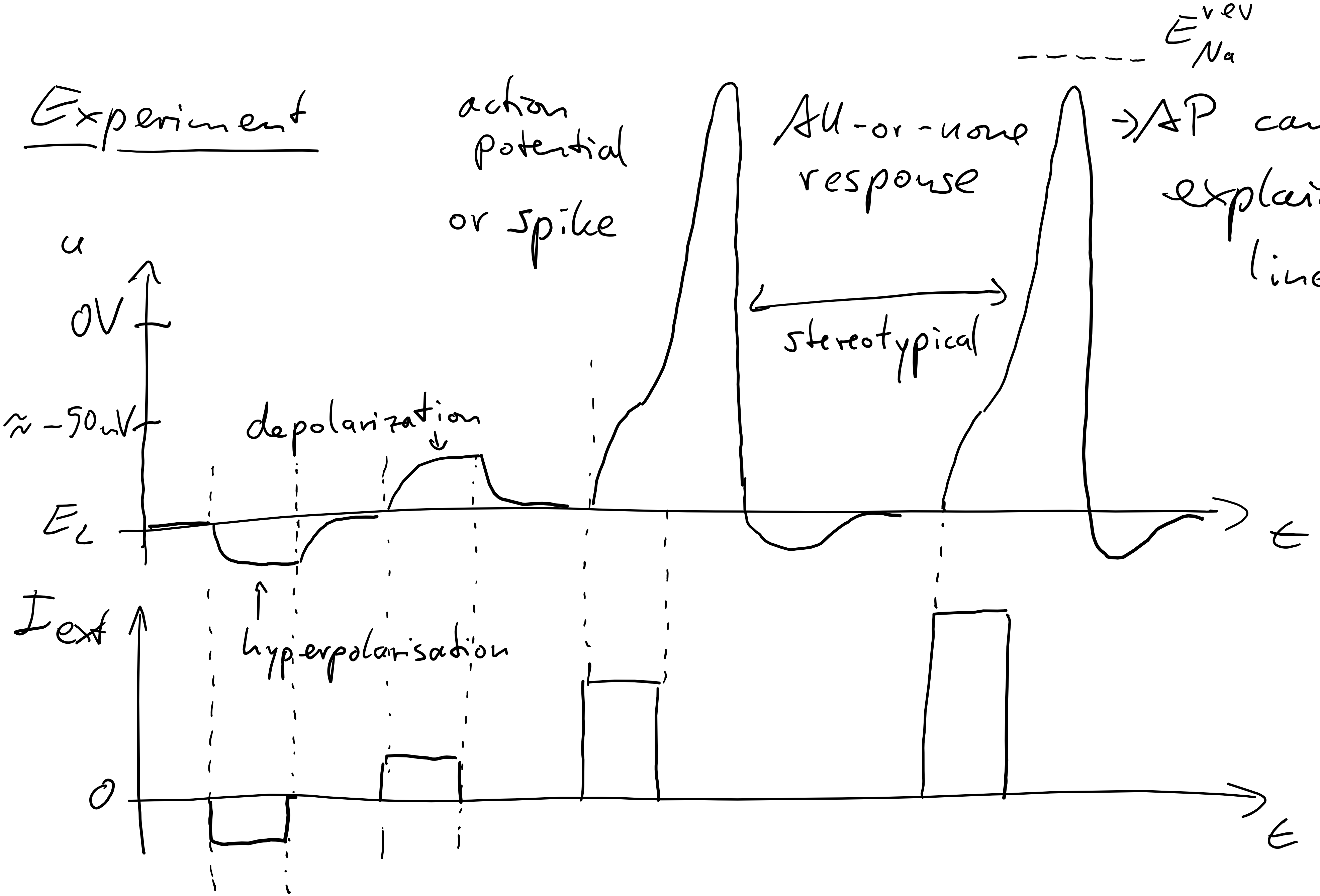
Neuronal exitability and action potentials

Experiment



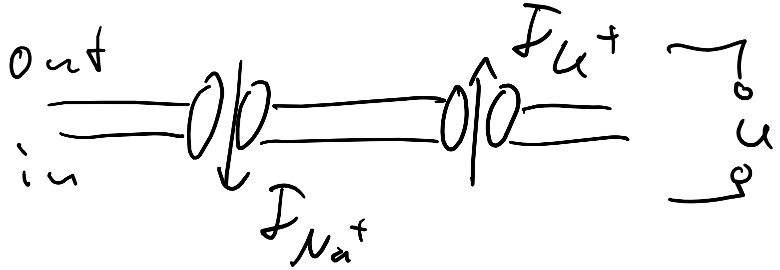
→ AP can not be explained by a linear ODE !

Solution: Voltage-gated Ion channels $g_x = g_x(u,t)$

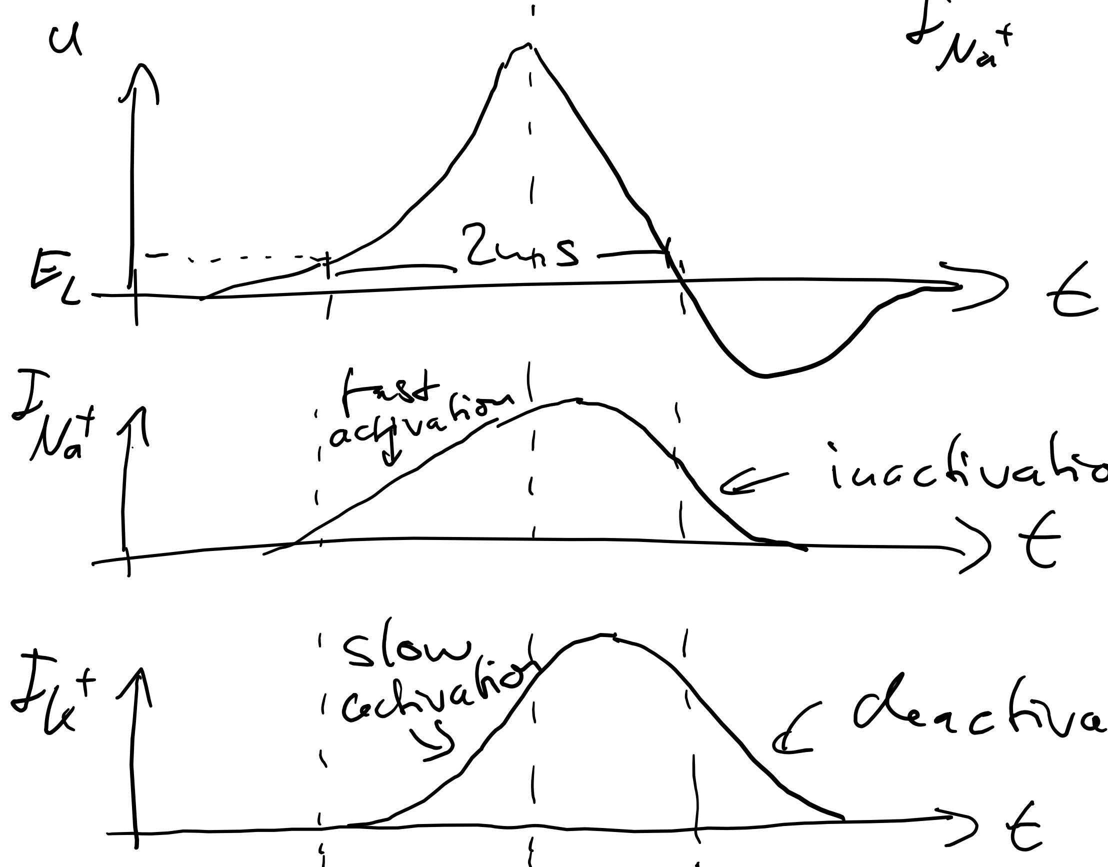
→ Especially in the membrane of

- the axon (the „output“ of the neuron)
- synapses (later in the lecture)

Experiment



opposite direction of NaK pump



inactivation (≠ deactivation)

deactivation

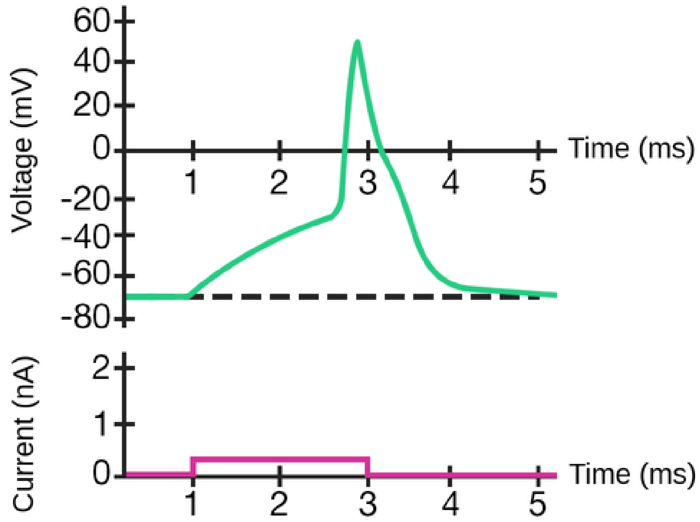
How can I_{Na} and I_K be measured (basic electrophysiology): „Voltage-Clamp“ by Kenneth Cole, 1947

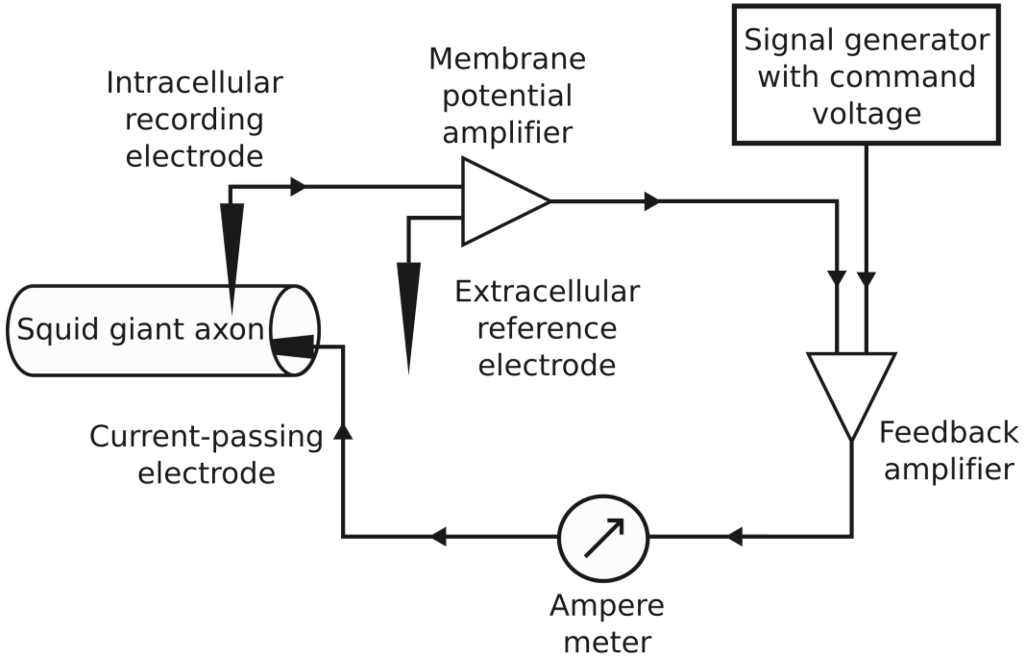
Inject a command current $I_c(t)$ into the cell such that the membrane potential u(t) follows a desired voltage trace (using fast electronics)

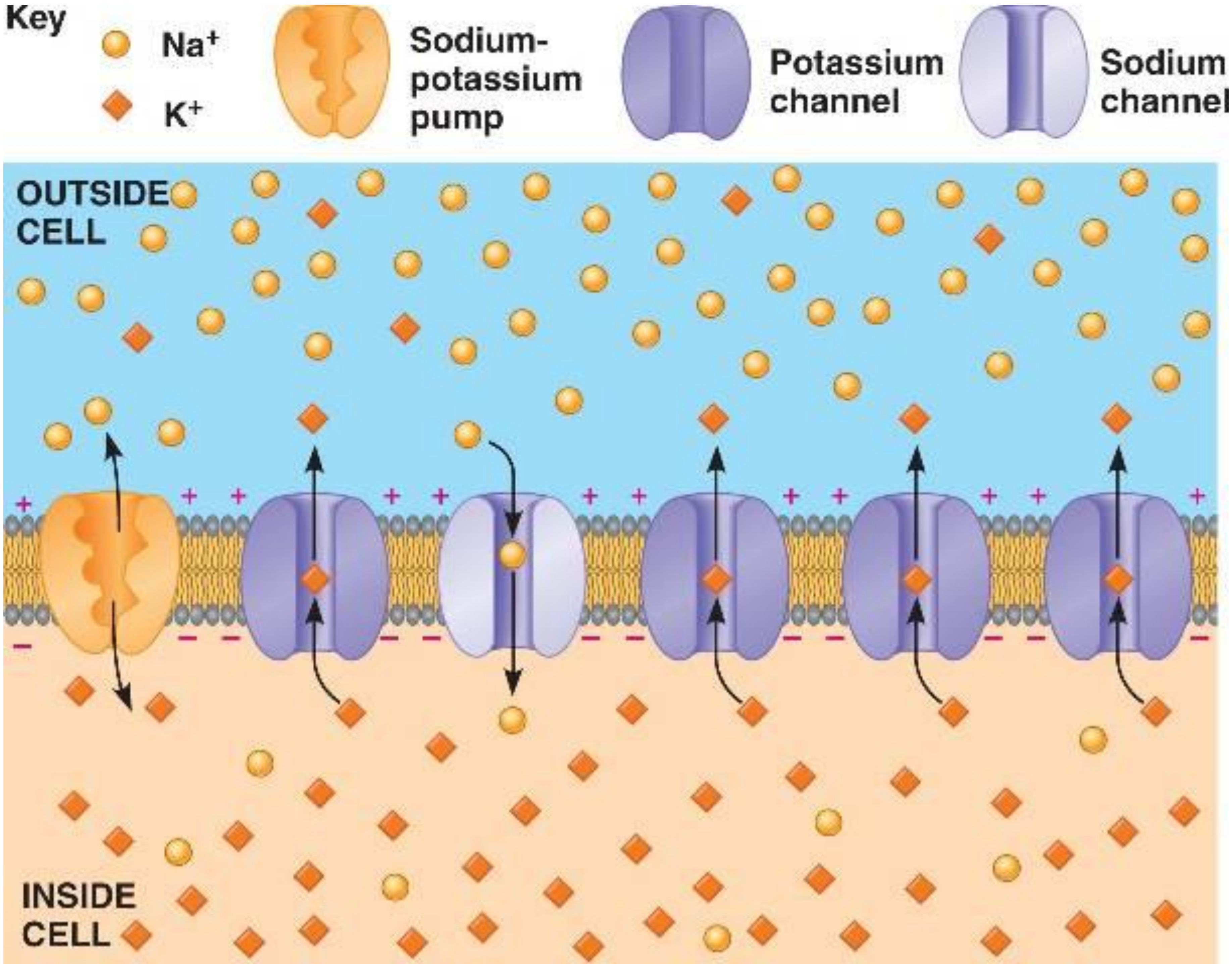
For the action potential:

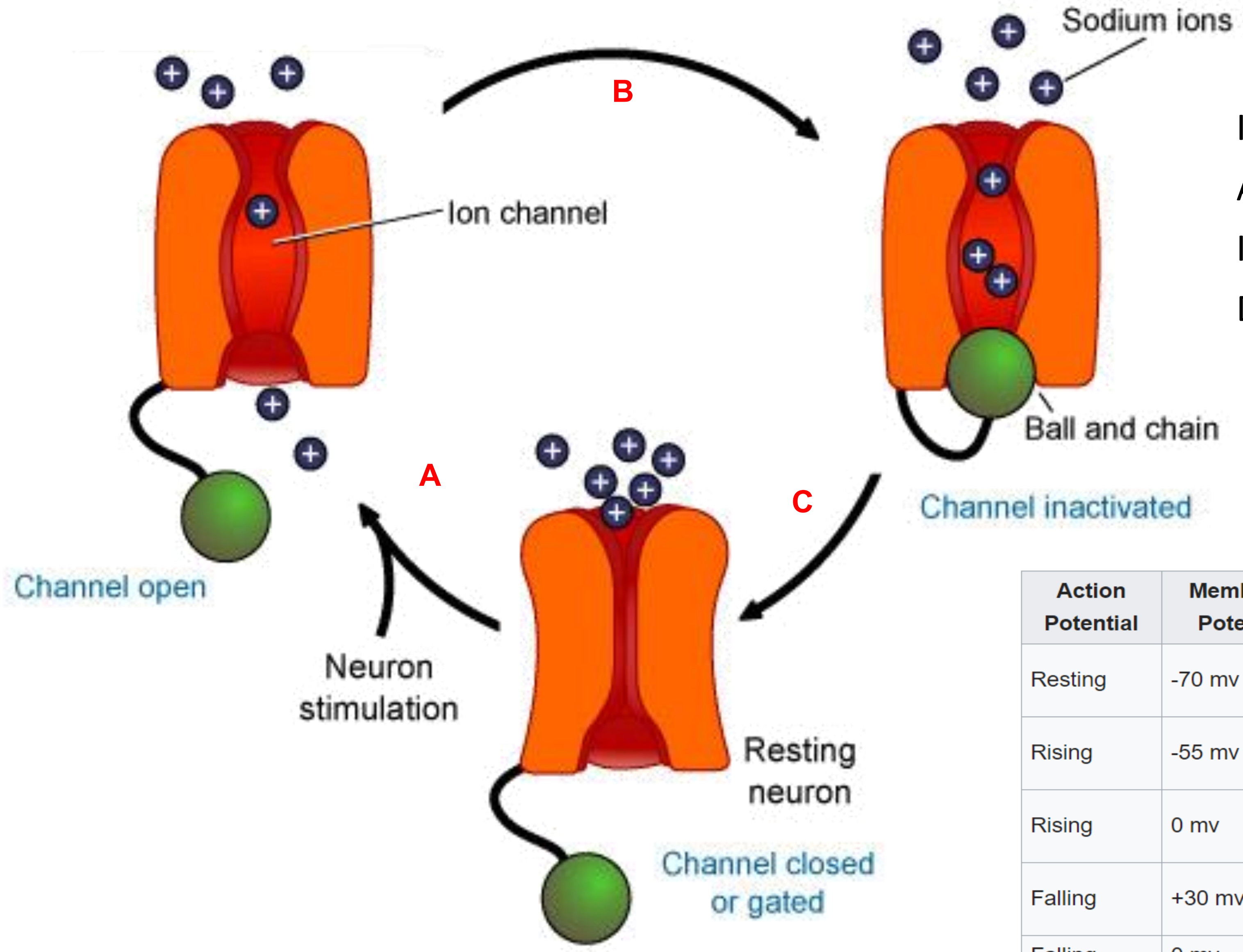
1. For the target trace u(t)=„action potential“, we expect $I_c(t)=0$
2. Block certain voltage-gated ion channels, eg. Na^+ channel via pharmacological blocker TTX
   → The required current $I_c(t)$ substitutes (= mirrors) I_{Na}^+
3. Calculate $g_{Na}(u,t)$ from $I_{Na} = g_{Na}(E_{Na}^{rev} – u)$ with the Nernst reversal potential E_{Na}^{rev}
   (the reversal potential is the voltage across the ion channel at which the current direction reverses:
   $u < E^{rev}$: $I < 0$
   $u > E^{rev}$: $I > 0$)

u(I) for action potential without blockers






Key
Na+
K+
Sodium-potassium pump
Potassium channel
Sodium channel
OUTSIDE CELL
INSIDE CELL



Ion-Channel has three distintive states:

Activated: channel is open

Inactivated: closed, can not be opened (activated)

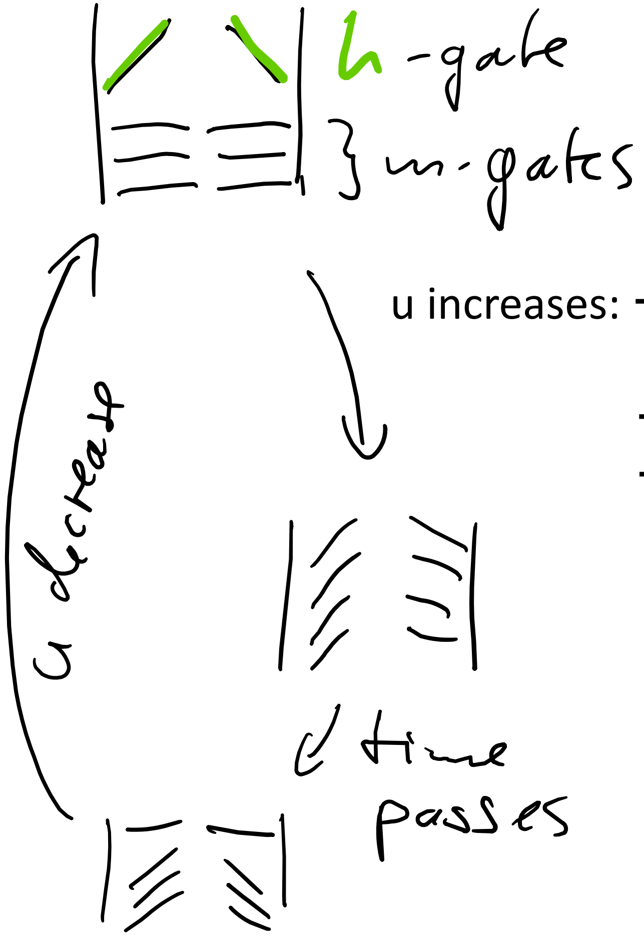
Deactivated: closed, but ready to be activated

Action Potential	Membrane Potential	Target Potential	Gate's Target State	Neuron's Target State
Resting	-70 mv	-55 mv	Deactivated → Activated	Polarized
Rising	-55 mv	0 mv	Activated	Polarized → Depolarized
Rising	0 mv	+30 mv	Activated → Inactivated	Depolarized
Falling	+30 mv	0 mv	Inactivated	Depolarized → Repolarized
Falling	0 mv	-70 mv	Inactivated	Repolarized
Undershot	-70 mv	-75 mv	Inactivated → Deactivated	Repolarized → Hyperpolarized
Rebounding	-75 mv	-70 mv	Deactivated	Hyperpolarized → Polarized

A functional model of voltage-gated channels

Sodium channels:

- 4 gates: 1h, 3m
- stochastic opening depends on voltage u



probability of being open: h
p of all 3 being open: m^3

u increases: → fraction of open channels:

m^3h

→ conductance: $g_{Na}^{eff} = g_{Na}\, m^3h$

→ current: $I_{na} = g_{Na} m^3 h(E_{na}^{rev} - u)$

Gating variables m, h that obey:

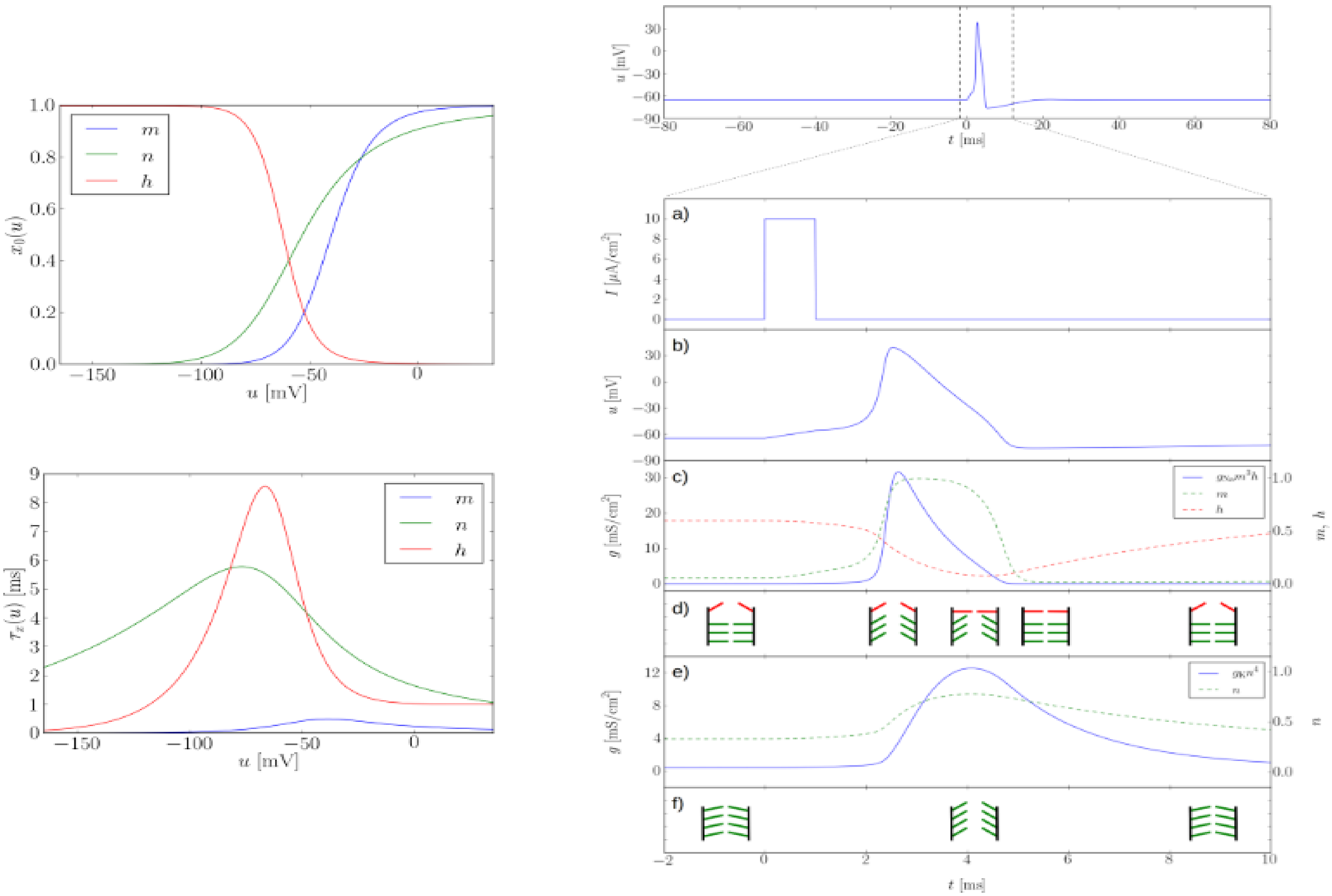
$$\frac{dm}{dt} = \frac{1}{\tau_m(u)}(m_0 - m) \quad m = m(u,t)$$

$$\frac{dh}{dt} = \frac{1}{\tau_h(u)}(h_0 - h) \quad h = h(u,t)$$

Postassium channels:

- 4 identical gates

$$\frac{dn}{dt} = \frac{1}{\tau_n(u)}(n_0 - n) \quad n = n(u,t)$$



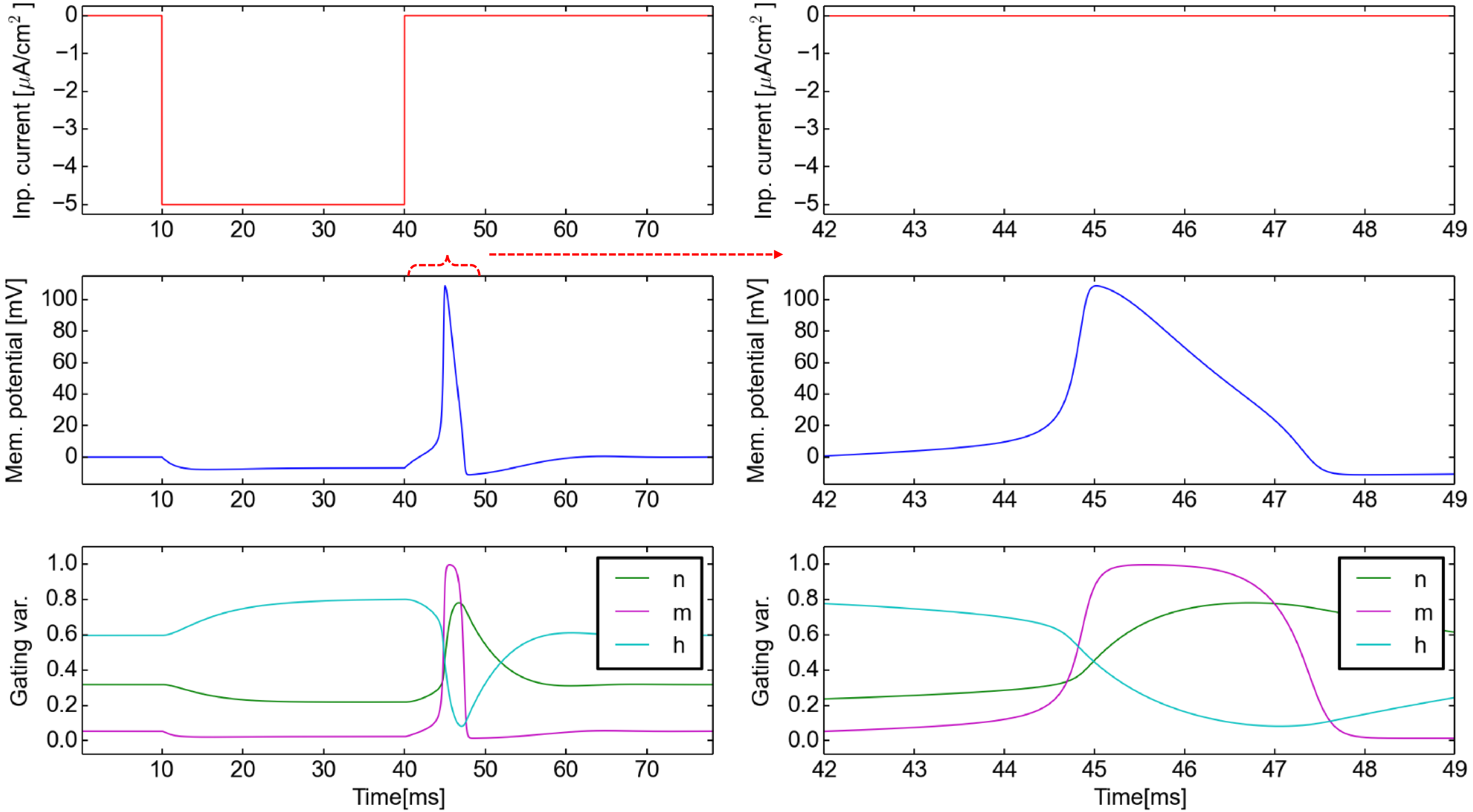
m
n
h
$x_0(u)$
u [mV]
$\tau_x(u)$ [ms]
u [mV]
u [mV]
t [ms]
a)
I [μA/cm^2]
b)
u [mV]
c)
g [mS/cm^2]
m, h
d)
e)
g [mS/cm^2]
n
f)
t [ms]

The Hodgkin-Huxley model

Developed by Alan Hodgkin and Andrew Huxley around 1952 → Nobel Prize 1963

$$C_m \dot{u} = g_L (E_L - u) + g_{Na} m^3 h (E_{Na} - u) + g_K n^4 (E_K - u) + I^{ext}$$

$$\dot{x} = \frac{1}{\tau_x(u)} \left(x_0(u) - x \right) \qquad x \in \{m, n, h\}$$

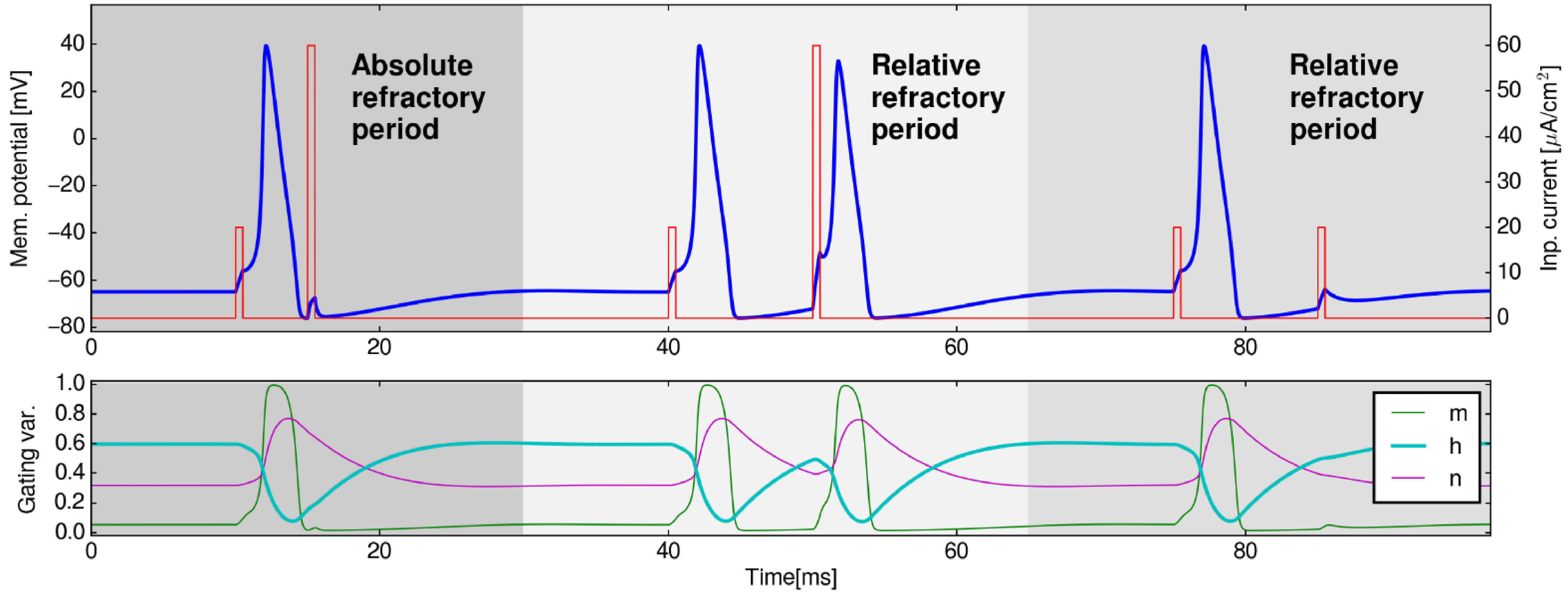

Inp. current [μA/cm^2]
0
−1
−2
−3
−4
−5
10
20
30
40
50
60
70
42
43
44
45
46
47
48
49
Mem. potential [mV]
100
80
60
40
20
0
Gating var.
1.0
0.8
0.6
0.4
0.2
0.0
n
m
h
Time[ms]

Three interesting properties of the Hodgkin-Huxley model

1. Is there a firing threshold?
   → rheobase := smallest current to elicit a spike
   → firing depends on the **amplitude** of the stimulus and at the **speed** at which it changes!
2. Post-inhibitory rebound spikes
   → An abruptly ending inhibitory stimulus can elicit a spike
3. Resonance
   → Appropriately spaced „subthreshold“ stimuli can cause a spike!

Response properties of the HH model

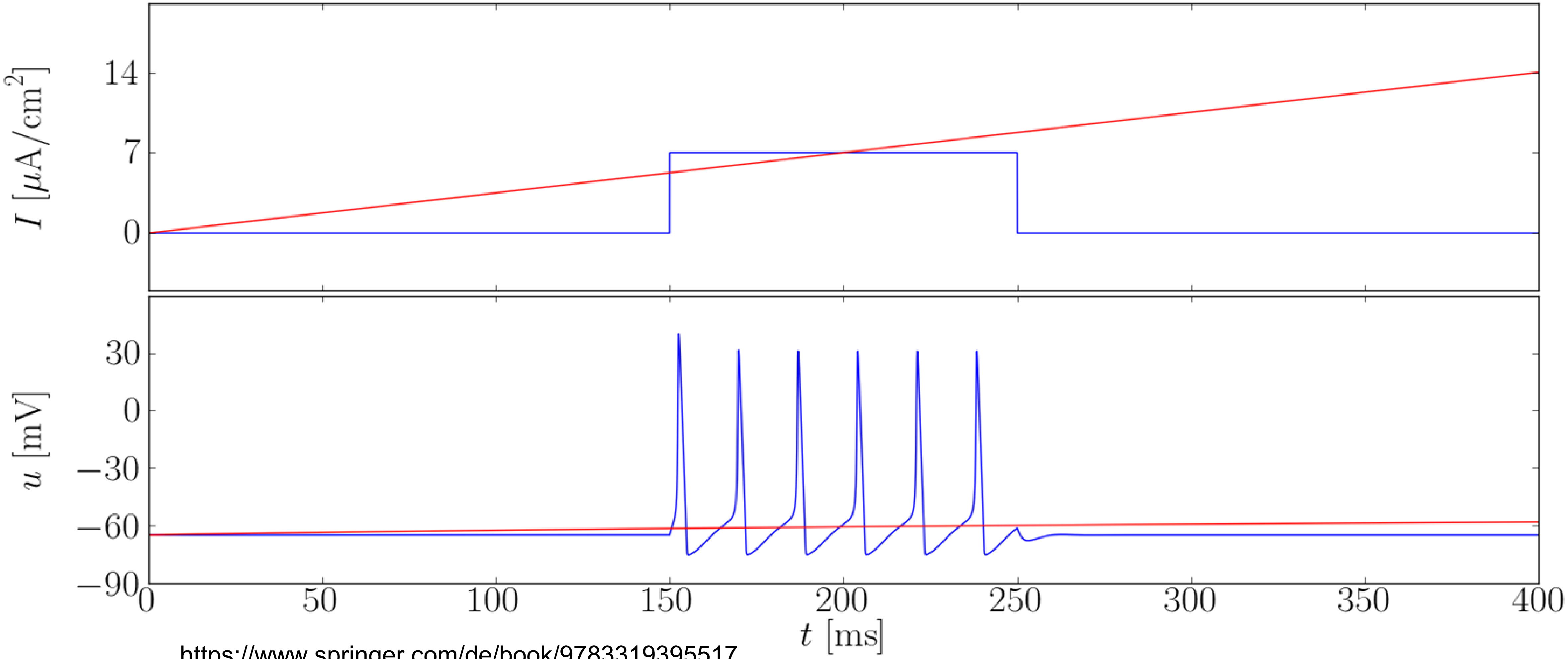
1. Refractory period, i.e. period where the neuron can no spike again, following each spike
   - Absolute refractory period: while the h-gates are still closed
     → inactivation, no new Na^+ currents
   - Relative refractory period: gradual recovery of excitability as the gating variables decay to their resting states



<u>Response properties of the HH model</u>

no simple rheobase (= smallest constant to elicit a spike)

Searching for the rheobase:



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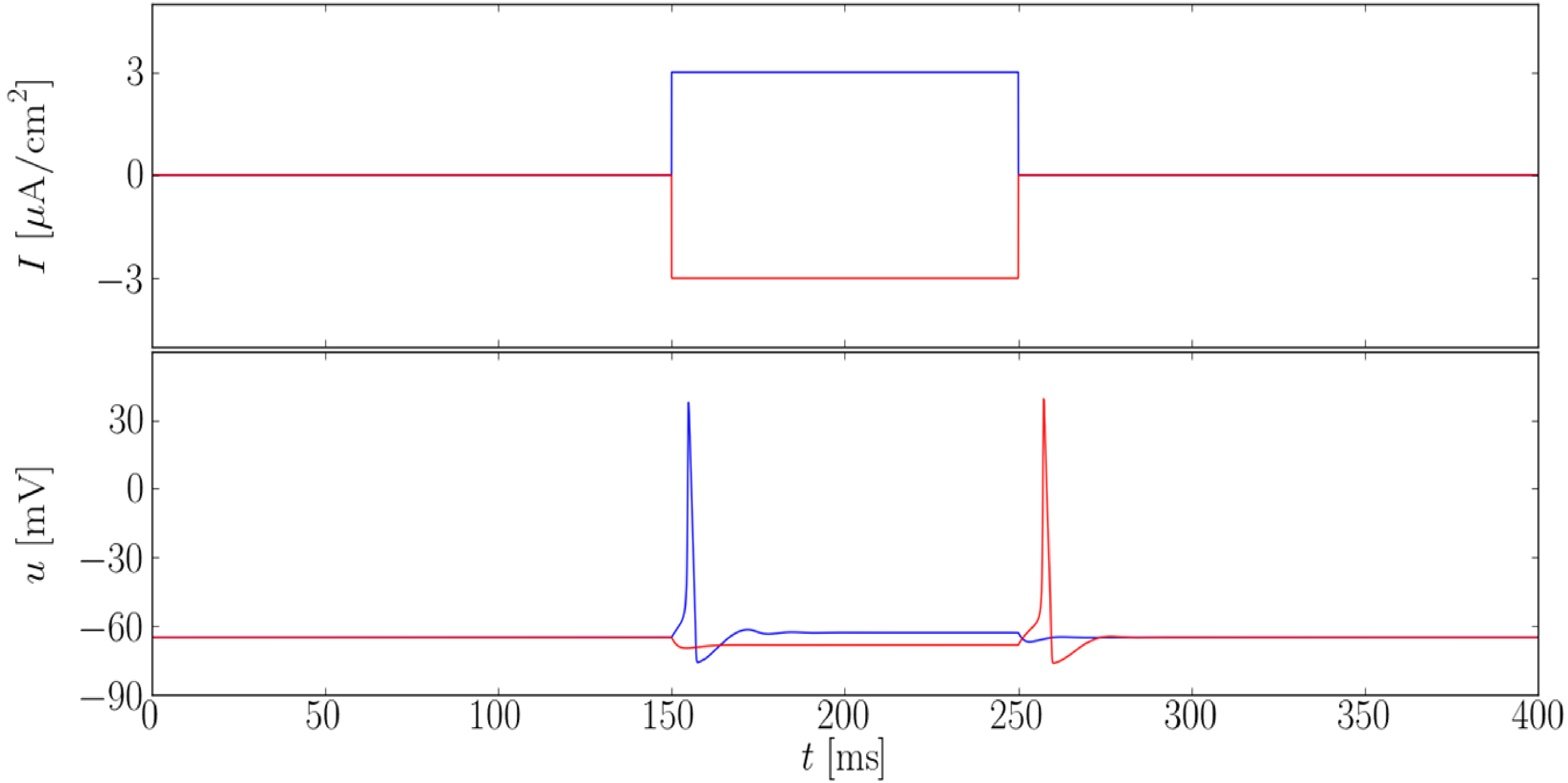
Inhibitory rebound



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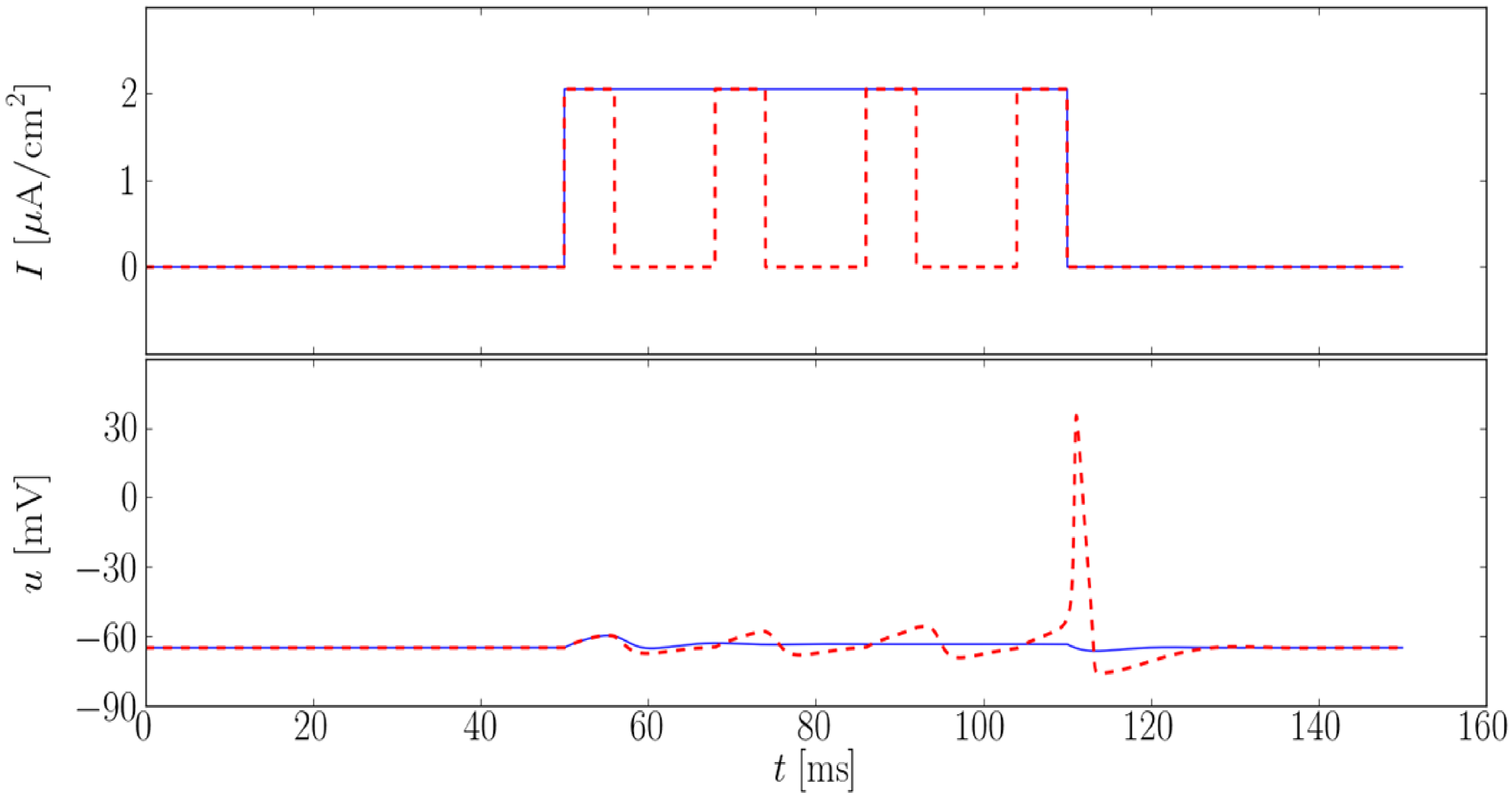
Resonance phenomenon



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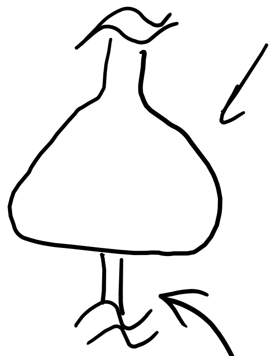
Summary so far



1. "Passive" membrane

   Na^+, K^+, … channels

   → resting potential

   → leaky integration with $\tau_m = \dfrac{C_m}{g_L}$

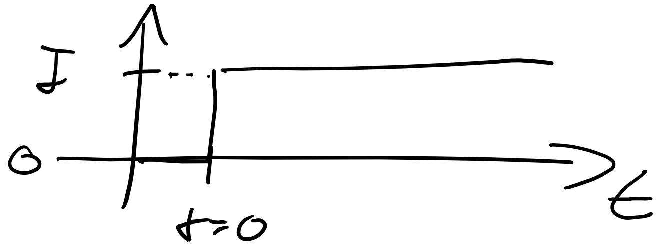
2. "active" axon

   Gating variables $m(t)$, $h(t)$, $n(t)$

   → Hodgkin-Huxley model with action potentials

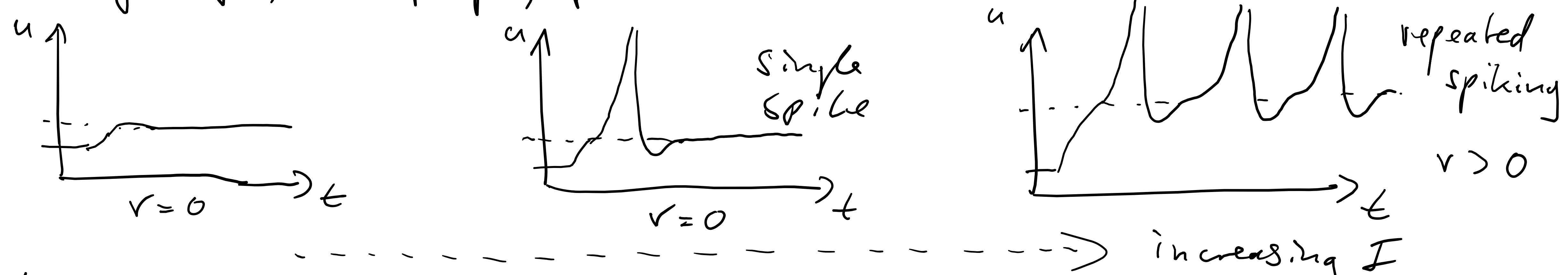
Response to simple step currents : activation function

Def: activation function

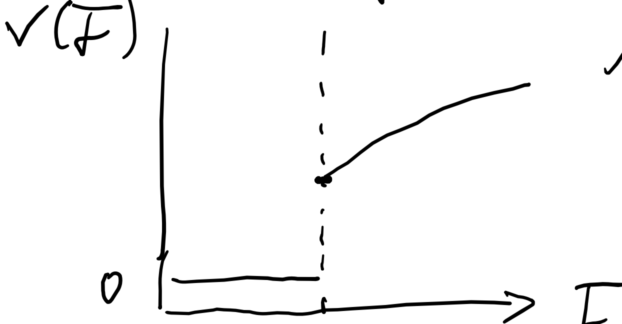


$$r(I) = \lim_{T\to\infty} \frac{\#\text{spikes in } [0,T)}{T}$$

aka. gain fct, transfer fct, f-I curve

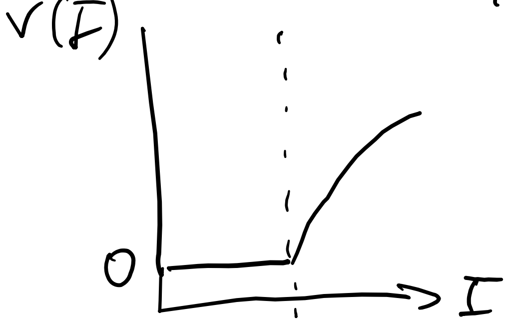


Activation function



Neurons with a sudden transition to repetitive firing with $r>0$ are called "Type II" neurons.

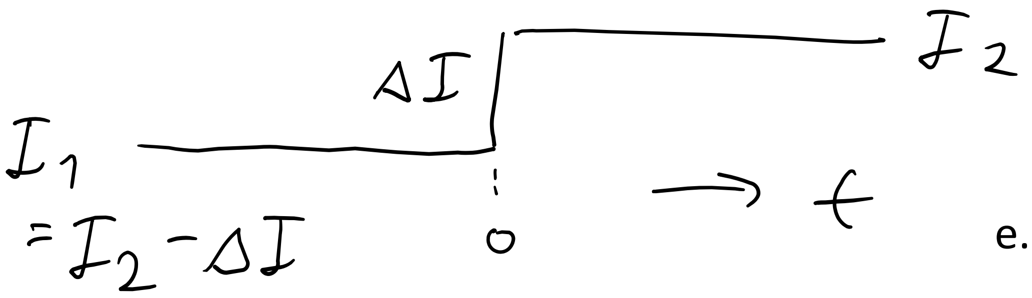
· There are also "Type I" neurons



Type I neurons have a continuous activation function

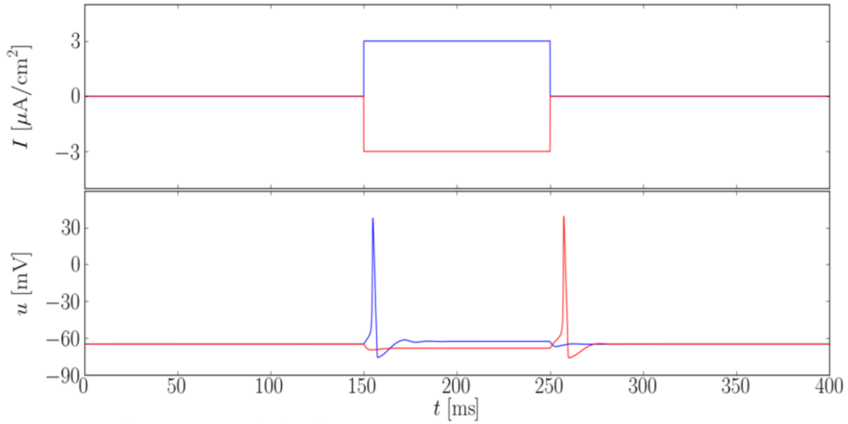
In biology, we find both Type I and Type II neurons.

We are normally interested in step currents beyond the activation function (i.e. including history)



e.g. post-inhibitory rebound spike

Inhibitory rebound



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$I(t)=I_2 - \Delta I + \Delta I\,\Theta(t)$ with the Heaviside step fct: $\Theta(t) = 1$ for $t\geq 0$; 0 for $t<0$

→ two independent parameters (I_2, ΔI)

the HH model consists of 5 coupled ordinary differential equations (ODEs)

- hard to analyse
- slow to simulate (small timestep Δt required)

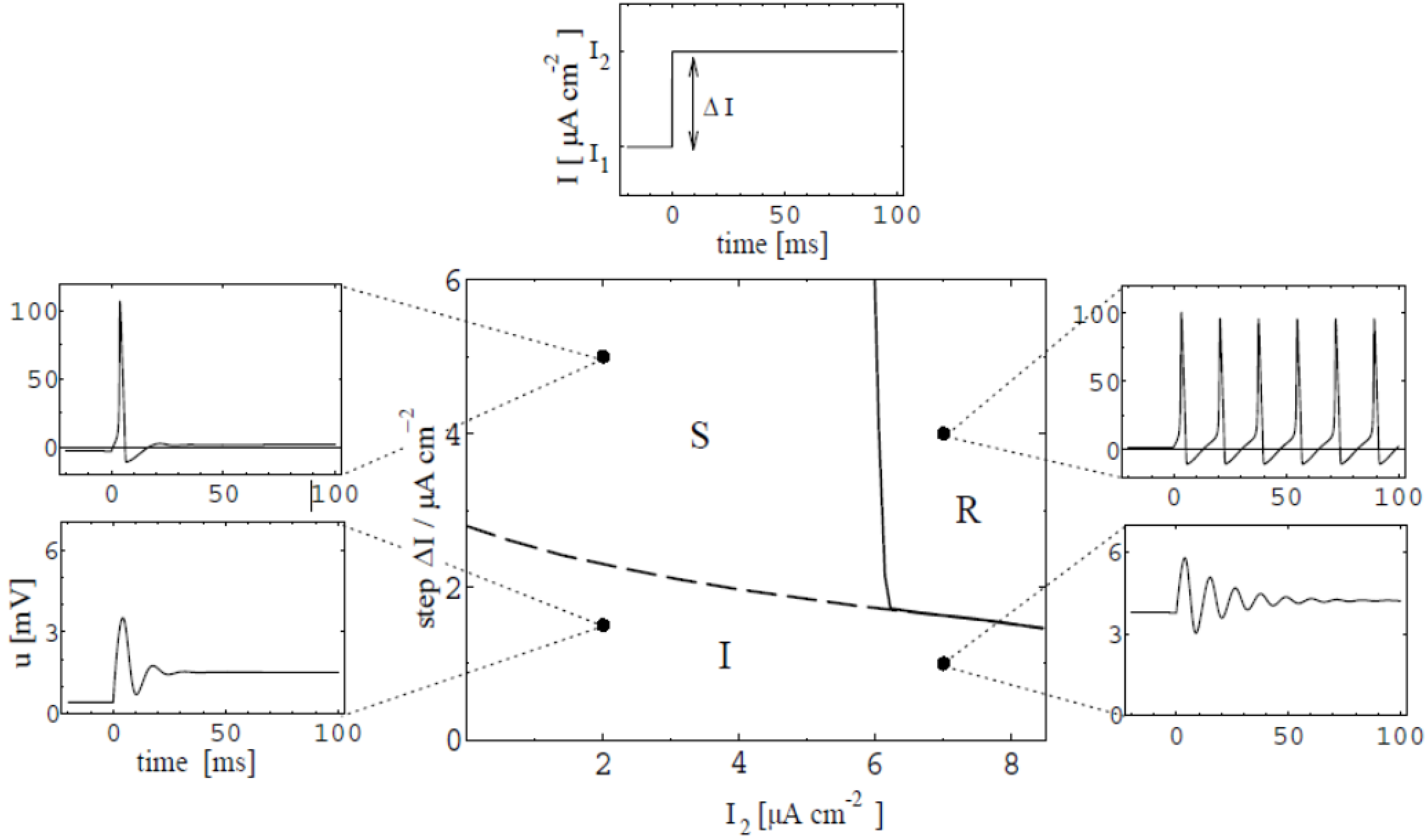
→ **simplified neuron models are desirable !**

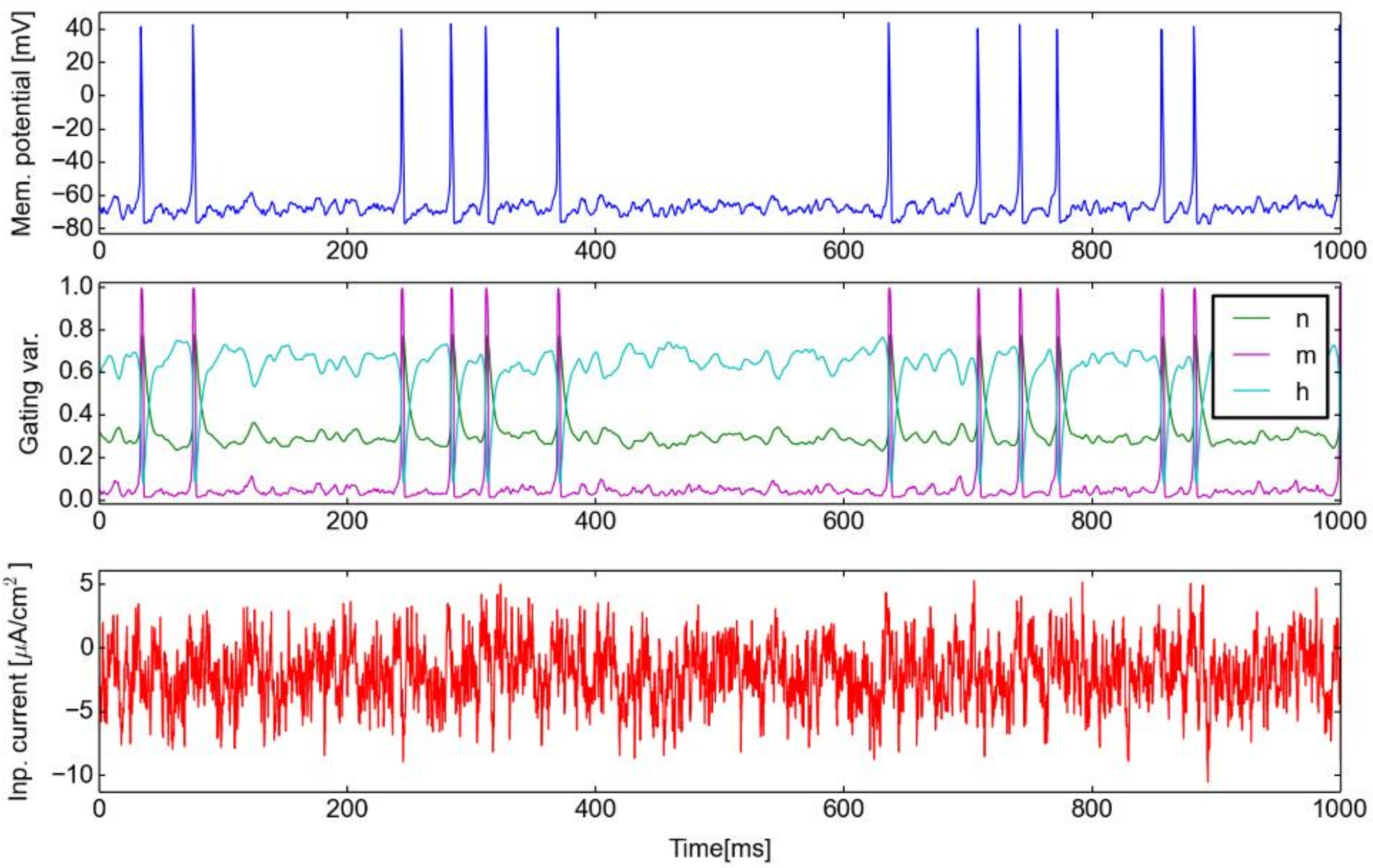
Hodgkin-Huxley model : Phase diagram for stimulation with a step current.

I : inactive

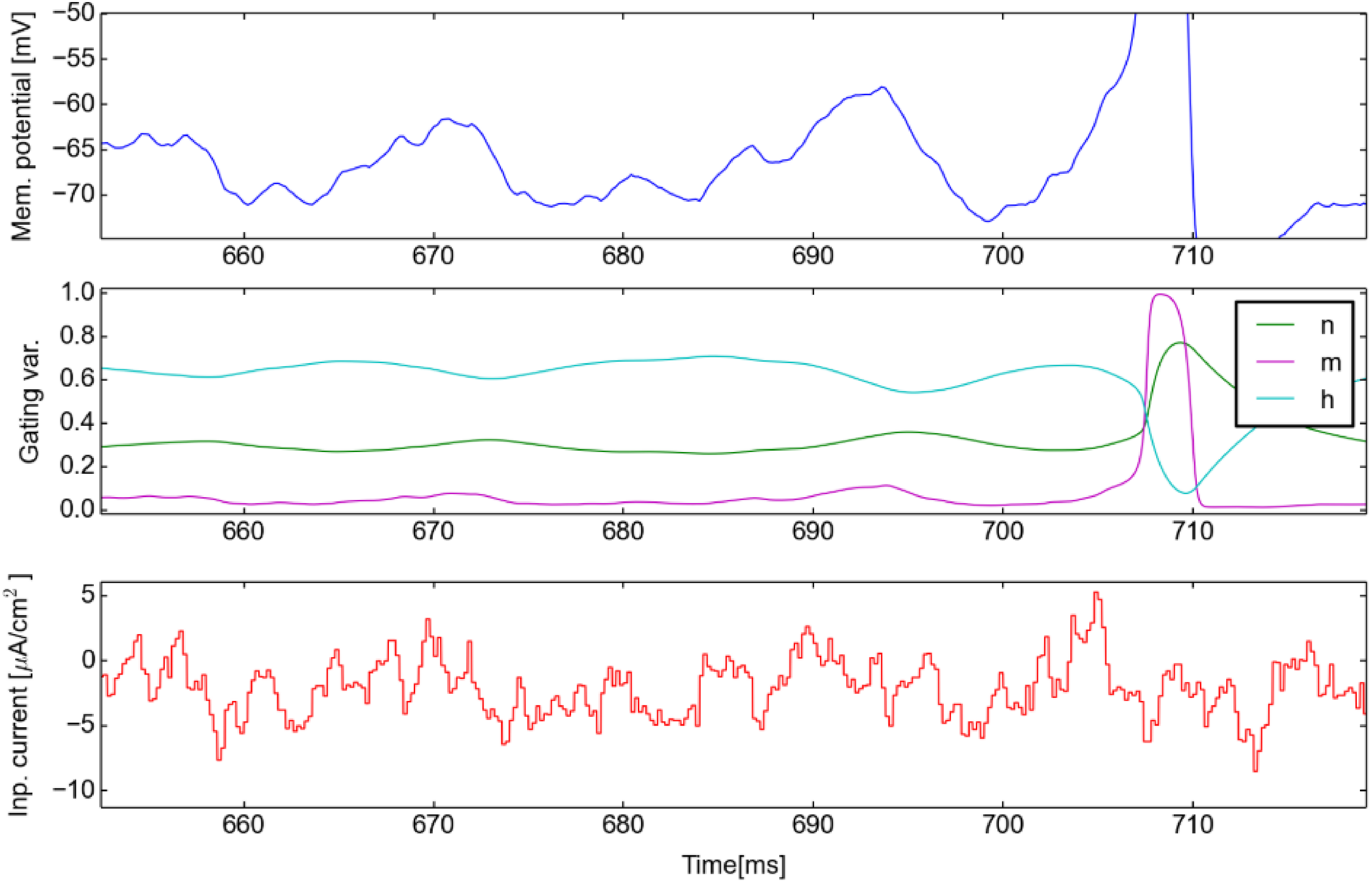
S : single

R : repetitive





Mem. potential [mV]
40
20
0
−20
−40
−60
−80
0
200
400
600
800
1000
Gating var.
1.0
0.8
0.6
0.4
0.2
0.0
n
m
h
Inp. current [μA/cm^2]
5
0
−5
−10
Time[ms]



Mem. potential [mV]
−50
−55
−60
−65
−70
660
670
680
690
700
710
Gating var.
1.0
0.8
0.6
0.4
0.2
0.0
n
m
h
Inp. current [μA/cm²]
5
0
−5
−10
Time[ms]